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Nambur, Pedakakani (M), Guntur (Dt) - 522508
Department of CSE - Artificial Intelligence & Machine Learning

MAJOR PROJECT REGISTRATION FORM

Date:31-07-2026

Project Category:	☑ Research oriented ☐ Application oriented ☐ Product oriented ☐ Other (please specify) Specify if others:	Batch ID:	**C6**

.

Title of the Project:

MedIntel: AI-Powered Smart Hospital Appointment and Clinical Decision Support System

Abstract

Many hospitals still use manual or partially digital systems to manage outpatient appointments and patient records, resulting in long waiting times, repeated paperwork, poor communication among healthcare staff, and unnecessary delays. A common issue is that patients must explain the same health problem multiple times—first to the assistant doctor or nurse during the initial examination and then again to the consulting doctor. This reduces consultation efficiency and may lead to incomplete or inconsistent medical records.

This project proposes **MedIntel**, an **AI-Powered Smart Hospital Appointment and Clinical Decision Support System**, a web-based platform that integrates Artificial Intelligence (AI), Machine Learning (ML), and Natural Language Processing (NLP) to improve outpatient healthcare services. The system enables patients to register, select hospitals and doctors, book appointments, enter symptoms, medical history, allergies, current medications, and upload previous medical reports before visiting the hospital. During the initial examination, healthcare staff record patient complaints, vital signs, and observations into the system, allowing consulting doctors to instantly access complete patient information without requiring patients to repeat their medical history.

The proposed system incorporates several AI-powered modules, including an **AI Symptom Classification Model** for department recommendation, a **Smart Doctor Recommendation System**, an **Intelligent Appointment Scheduling System**, and a **Patient Priority Prediction Model** that classifies patients into Routine, Moderate, High, or Emergency categories based on symptoms, age, medical history, and vital signs. To support clinical decision-making, the system also includes an AI Similar Clinical Case Retrieval Module, which retrieves anonymized historical cases with similar symptoms and treatment outcomes, an **AI Clinical Summary Generator** that automatically prepares concise patient summaries, an **AI Health Trend Analyzer** that analyzes previous medical records to identify trends in health indicators such as blood pressure, blood sugar, cholesterol, and BMI, and an **OCR-based Medical Report Analyzer** that extracts information from uploaded reports to reduce manual data entry.

Overall, MedIntel provides a secure, scalable, and intelligent healthcare platform that efficiently manages **Electronic Medical Records** (EMR), prescriptions, laboratory reports, appointment history, and follow-up details while reducing administrative workload, improving communication between patients and healthcare professionals, supporting better clinical decision-making, and enhancing the overall quality and efficiency of outpatient healthcare services.

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Project Team Members

S.No.	Regd. No.	Name of the Student	Signature
1.	23BQ1A42H6	THOTA ROHINI	
2.	23BQ1A42D5	PATNAM VENKATA NIHITHA	
3.	23BQ1A42I8	YARLAGADDA MAHESH CHANDRA BOSE	
4.	24BQ5A4218	VELLATURI JYOSHNA SRI	

Project Guide

Name: DR. M. SRINIVASA RAO

Signature: